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# MIRROR: Learning from the Other View for Multi-Modal Reasoning

*Modality-Informed Reciprocal Reasoning Optimization Trains Vision-Language Models to Transfer Strengths Across Text, Diagram, and Combined Views of the Same Problem*

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Vision-language models fail at visual reasoning even on geometry problems that have mathematically equivalent text, diagram, and combined views. MIRROR, from researchers at USC and CMU, exploits this inconsistency: rather than treating it as noise, the method identifies which view a model currently solves best and uses that signal to lift performance on the harder view through reciprocal cross-modal reward shaping during RL training. On ODA-Data, a new paired multimodal geometry benchmark, MIRROR improves image-dominant reasoning by 17 points over a strong baseline.

## AT A GLANCE

**Problem:** VLMs solve the same geometry problem in text but fail on the diagram, revealing complementary reasoning gaps across modalities that standard post-training ignores.

**Why it matters:** Closing modality gaps enables VLMs to reason reliably regardless of how a problem is presented—critical for real-world deployment where input format is unpredictable.

**Method:** MIRROR evaluates each view per-problem, identifies the best-performing view, and shapes rewards on weaker views toward the best-view distribution via RL.

**Result:** +17pp on image-dominant and +9pp on combined view vs. GRPO baseline on ODA-Data geometry; consistent gains on three external geometry benchmarks.

**Evidence:** Ablating cross-modal shaping loses 9.4pp image accuracy; scrambled skills confirm content drives improvement.

Vision-language models are post-trained on massive image-text pairs and achieve strong results on captioning, VQA, and chart understanding. Yet on mathematical visual reasoning—where the model must interpret a geometric diagram and apply deductive inference—performance lags substantially behind text-only reasoning.

The standard explanation is that vision encoding is weaker than language encoding. MIRROR’s key insight is that this framing misses a more specific phenomenon: the *same problem* in a different modality often elicits dramatically different behavior from the same model. A model that easily solves a geometry problem from a text description may completely fail on the same problem presented as a diagram—not because it lacks geometric knowledge, but because it fails to activate that knowledge from visual input.

## Why Existing Approaches Fall Short

Standard VLM post-training mixes image-text pairs and optimizes a single objective (SFT or RLHF) over all examples, treating modality as a property of the input rather than an independent factor in reasoning quality. This approach:

- **Averages over view-dependent performance:** A model that excels on text but fails on diagrams receives the same training gradient regardless of which view it was shown.
- **Does not exploit complementary signals:** The model’s success on text implicitly encodes geometric reasoning that could guide improvement on diagrams—but standard training does not route this signal across modalities.
- **Ignores asymmetric capability:** Per-view performance profiles differ across problems; a fixed training mix cannot leverage which view each problem is easiest from.

## Core Mechanism

MIRROR introduces a per-problem view evaluation step before each RL update. For each training problem  $p$ :

1. Evaluate the current policy on all views  $\mathcal{V} = \{T, I, T+I\}$  to obtain per-view rewards  $r(p, v)$ .
2. Identify the best view:  $v^*(p) = \arg \max_v r(p, v)$ .
3. Compute augmented rewards for weaker views that include a cross-modal alignment term encouraging the weaker-view distribution to match the best-view distribution.

The policy update uses GRPO with the augmented rewards, simultaneously training on all views. The reciprocal structure ensures that improvement on the text view can

## The Big Picture

feed into the image view and vice versa.

### Technical Formulation

Let  $\pi_\theta$  be the policy and  $a$  the correct answer. The base reward is  $r(p, v) = \mathbf{1}[\pi_\theta(p_v) = a]$ . The MIRROR-augmented reward for view  $v \neq v^*(p)$  is:

$$\tilde{r}(p, v) = r(p, v) + \lambda \cdot \text{KL}\left(\pi_\theta(\cdot \mid p_{v^*}) \parallel \pi_\theta(\cdot \mid p_v)\right)^{-1}$$

The KL term penalizes divergence from the best-view response distribution, rewarding weaker views for matching the best-view reasoning trace. The inverse ensures the bonus is higher when the weaker view is already close to the best view (facilitating transfer) and smaller when the gap is large (avoiding overfit to a single view).

The full GRPO objective with MIRROR shaping:

$$\mathcal{L}_{\text{MIRROR}} = -\mathbb{E}_{p, v \sim \mathcal{D}} \left[ \tilde{r}(p, v) \frac{\pi_\theta(a_v \mid p_v)}{\pi_{\theta_{\text{old}}}(a_v \mid p_v)} - \beta \text{KL}(\pi_\theta \parallel \pi_{\text{ref}}) \right]$$

### Learning or Inference Procedure

Training uses ODA-Data for all views simultaneously. Each mini-batch contains all three views of the same problem, enabling within-batch comparison of view-dependent rewards. The  $\lambda$  coefficient governing cross-modal shaping strength is annealed from 0.5 to 0.1 over training to prevent the alignment term from dominating late-stage learning.

At inference, no change is needed: the model is queried with whichever view is available. The improvement is baked into the model weights.

### What the Guarantee Says

MIRROR does not guarantee cross-view performance parity—some problems may remain fundamentally harder in one view than another. What it provides is a systematic mechanism for reducing the gap, ensuring that the model’s best-view capabilities are not siloed and can transfer across the view boundary.

### Experimental Findings

#### ODA-Data test set:

| Method           | Text        | Image       | Combined    |
|------------------|-------------|-------------|-------------|
| LLaVA-1.6 (base) | 61.3        | 44.2        | 58.7        |
| + GRPO only      | 68.1        | 49.8        | 64.3        |
| + MIRROR         | <b>74.5</b> | <b>61.2</b> | <b>70.8</b> |

External geometry benchmarks (average over GeoQA,

UniGeo, Geometry3K): MIRROR +5.8pp over GRPO-only, +11.4pp over base.

The image-dominant view shows the largest absolute gain (+17pp over base), consistent with the hypothesis that image reasoning benefits most from cross-modal transfer from the textually stronger reasoning path.

### Ablations and Interpretation

**Remove cross-modal shaping** ( $\lambda = 0$ ): Image accuracy drops 9.4pp relative to full MIRROR. The model still improves over base from outcome RL, but the cross-view transfer is lost.

**Best view only** (train only on  $v^*$ , ignore weaker views): Image accuracy drops 14.1pp relative to MIRROR. Focusing only on the easiest view degrades weaker-view performance—the model over-specializes.

**Scrambled skills** (random KL target per problem): Image accuracy falls to 47.1%, below the GRPO-only baseline. The content of the cross-modal alignment target—not just the fine-tuning signal—drives the improvement.

**$\lambda$  annealing**: Fixed  $\lambda = 0.5$  degrades image accuracy by 3.2pp in late training, confirming that annealing is important for preventing the alignment term from dominating the RL signal.

### Reference

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